



Standards for National Health Claim Exchange (NHCX)

National Resource Centre for EHR Standards (NRCeS)

C-DAC Pune

- Background on Interoperability Standards implementation in ABDM
- About FHIR Standard
- FHIR Interoperability Paradigm
- Adoption of FHIR in ABDM
- Adoption of FHIR for NHCX
- Key Use cases
 - Profiling Requirements for NHCX
 - NHCX Standardization Approach
 - FHIR Resource Under Consideration for NHCX
 - Design Approach for NHCX
 - NHCX Profiles
 - Understanding the Structure of the Claim Bundle

Background - Health Information sharing over ABDM



- ABDM Implementation is based on the guidelines laid in the National Digital Health Blueprint (NDHB)
- Content & interoperability standards – FHIR, DICOM, SNOMED CT, LOINC, ICD
- A set of 8 essential and minimum class of health record artefacts for data capture in NDHB were recommended

Sr. No.	Information Record Purpose	Corresponding Resources in FHIR	
		Category	FHIR Resource
1	Patient Demographics Care Provider Details	Administration	Patient Practitioner Person
2	History, Problem & Diagnosis	Summary	Family Member History Condition Clinical Impression
3	Vitals, Results, Assessments (incl. Pregnancy, Death), Wellness parameters	Diagnostic	Observation Diagnostic Report
4	Adverse Event, Alert	Summary	Adverse Event Allergy Intolerance

5	Medication / Wellness Lifestyle / Diet / Vision	Medications	Medication Request Immunization
		Care	Nutrition Order Vision Prescription Care Plan Goal
6	Procedure	Care	Procedure
7	Admission / Discharge / Transfer / Order	Administration / Care	Appointment Encounter Episode of Care Service Request
8	Insurance	Financial	Coverage Coverage Eligibility Request / Response Claim / Claim Response

Fast Healthcare Interoperability Resource (FHIR)



- Health Data exchange standard developed and nurtured by HL7 International
- Internet-based approach to connecting different discrete elements
- Aims to build a set of base resources that, either by themselves or when combined, satisfy the majority of common use cases
- FHIR has around 150 resource types covering several categories including Entities, Clinical, and Financial types
- Representable in XML, JSON, or RDF formats
- Provides Restful CRUD specifications for common operations
- Extensible by design - allows usecase specific customization
- Terminology binding to popular code sets

Healthcare-related interoperability

Administration

Clinical Care

Research

Other

Scope

Why FHIR



Easy to develop: shallow learning curve and minimal custom tooling requirements

Easy to implement

Semantically Robust: can be mapped back to other standards

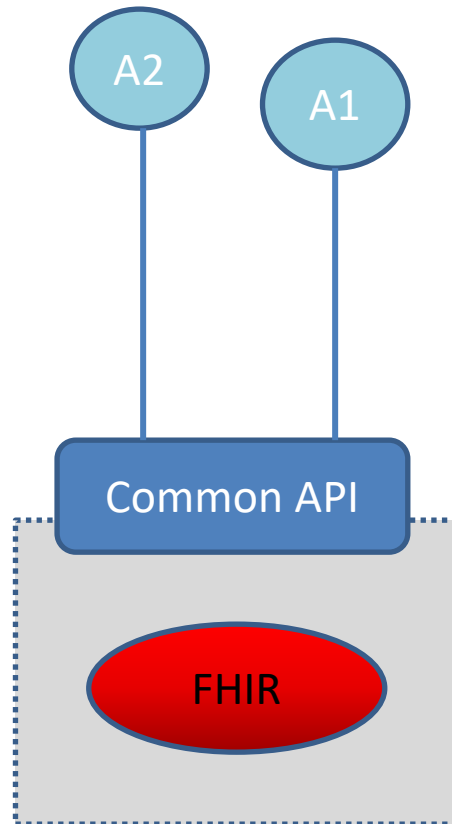
Implementer friendly: uses common tools and formats, and web-based technologies for the specification (HTTP, XML, JSON, etc.)

The artifacts/records can be validated electronically

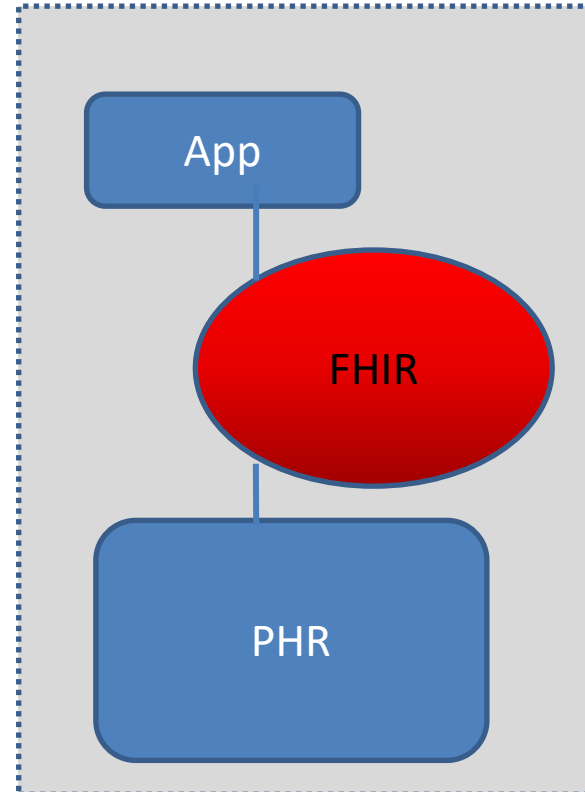
Open specification, multiple open-source implementations, vibrant community

Support human-readable serialization formats

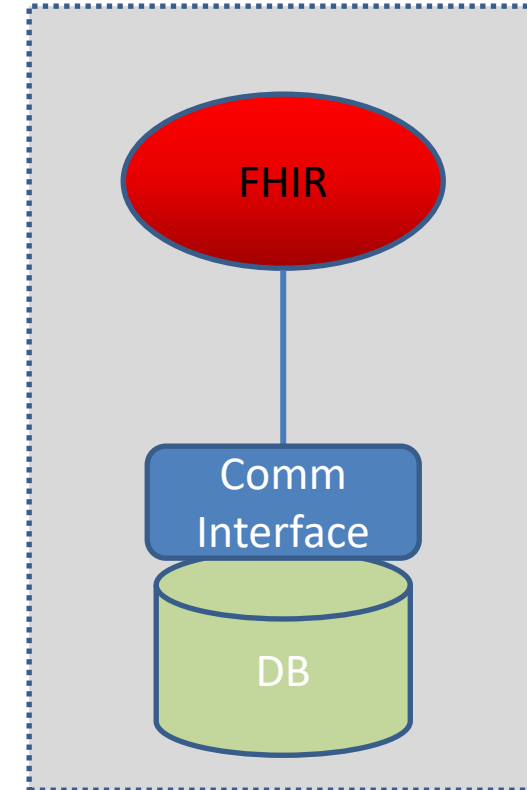
FHIR Usage Scenarios



FHIR object sharing



FHIR exchange

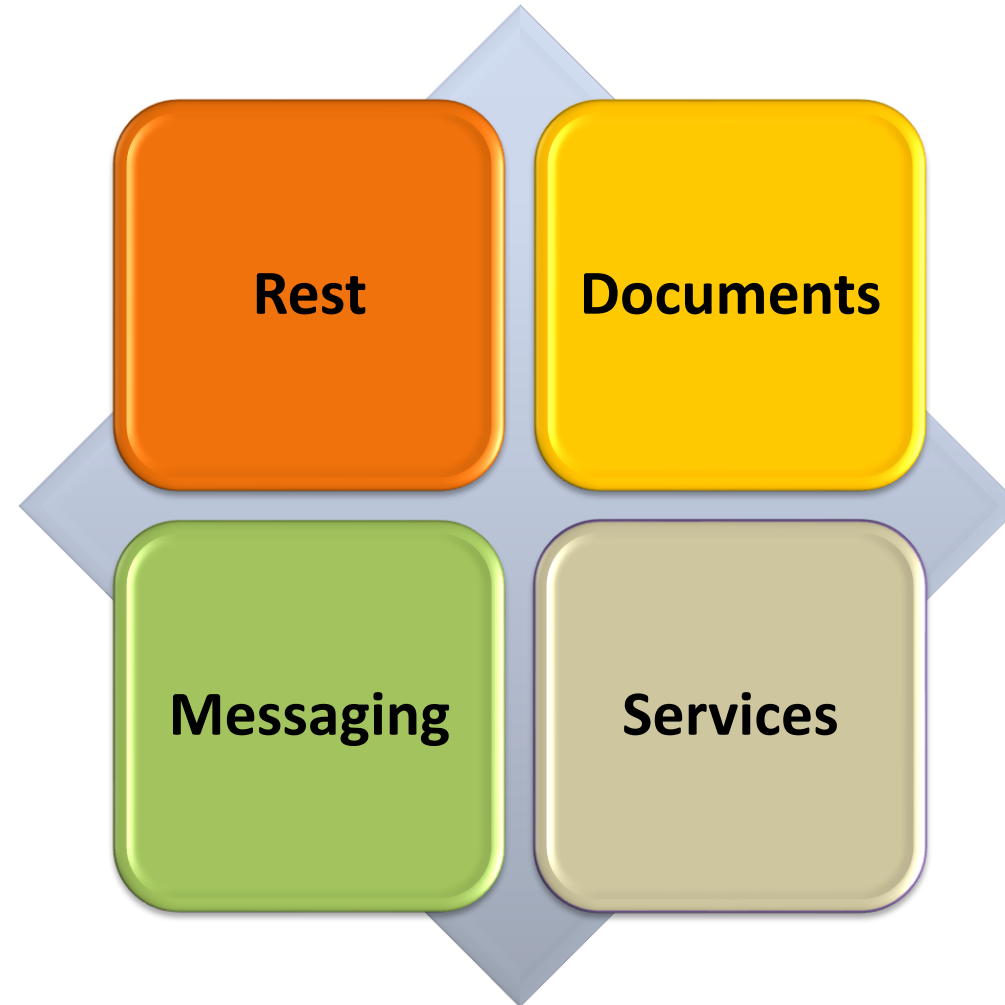


FHIR objects storage

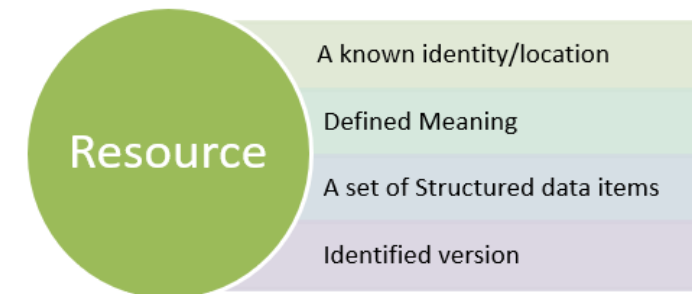
FHIR Interoperability Paradigms



FHIR supports interoperability via 4 paradigms



- [Resources](#) represent a unit of exchange of data that can be justifiable in interoperability
- A resource is made up of elements, each of which is a particular data type
- A resource has a known identity (a URL) by which it can be addressed
- Resources combine both computer processable and human-understandable data
 - Metadata to aid in searches and cataloging
 - Structural specification
 - Defined value sets for any enumerators
 - A display element for direct print or display of simplified text
 - Links to other resources
- Example of resources: [Patient](#); [Practitioner](#); [Observation](#); [Organization](#)



Key Parts of a Resource



```
<Patient xmlns="http://hl7.org/fhir">
  <id value="example-01"/>
  <meta>
    <versionId value="1"/>
    <lastUpdated value="2020-07-09T14:58:58.181+05:30"/>
    <profile value="http://nrces.in/ndhm/fhir/r4/StructureDefinition/Patient"/>
  </meta>
  <text>
    <status value="generated"/>
    <div xmlns="http://www.w3.org/1999/xhtml">Patient name - ABC, Age- 41 year, Gender- Male</div>
  </text>
  <extension url="https://nrces.in/ndhm/fhir/r4/StructureDefinition/someExtension">
    <valueCode value="someValue"/>
  </extension>
  <identifier>
    <type>
      <coding>
        <system value="http://terminology.hl7.org/CodeSystem/v2-0203"/>
        <code value="MR"/>
        <display value="Medical record number"/>
      </coding>
    </type>
    <system value="https://healthid.ndhm.gov.in"/>
    <value value="22-7225-4829-5255"/>
  </identifier>
  <name>
    <text value="ABC"/>
  </name>
  <telecom>
    <system value="phone"/>
    <value value="+919818512600"/>
    <use value="home"/>
  </telecom>
  <gender value="male"/>
  <birthDate value="1981-01-12"/>
</Patient>
```

Resource Identity
& Metadata

Human Readable Text

Extension with URL to Definition

Standard & Structured Data:

- Identifier
- Name
- Telecom
- Gender
- DOB

Resource list in FHIR





HomeGetting StartedDocumentationResourcesProfilesExtensionsOperationsTerminologies

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This page is part of the FHIR Specification (v4.0.1: R4 - Mixed Normative and STU). This is the current published version. For a full list of available versions, see the [Directory of published versions](#).

1.2 Resource Index

FHIR Infrastructure [Work Group](#)Maturity Level: N/ASTandards Status: Informative

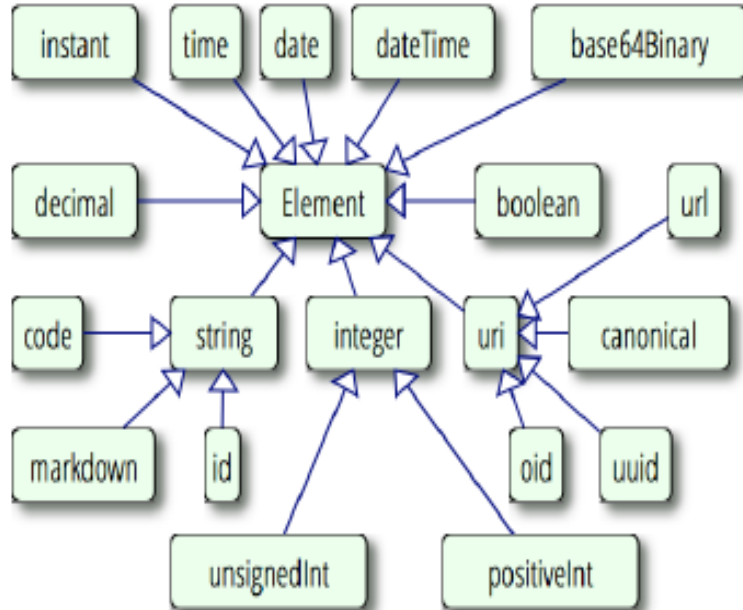
This page is provided to help find resources quickly. There is also a more [detailed classification, ontology, and description](#). For background to the layout on the layers in this page, see the [Architect's Overview](#). See also the abstract Base Resources [Resource](#) and [DomainResource](#).

CategorizedAlphabeticalR2 LayoutBy MaturitySecurity CategoryBy Standards StatusBy Committee

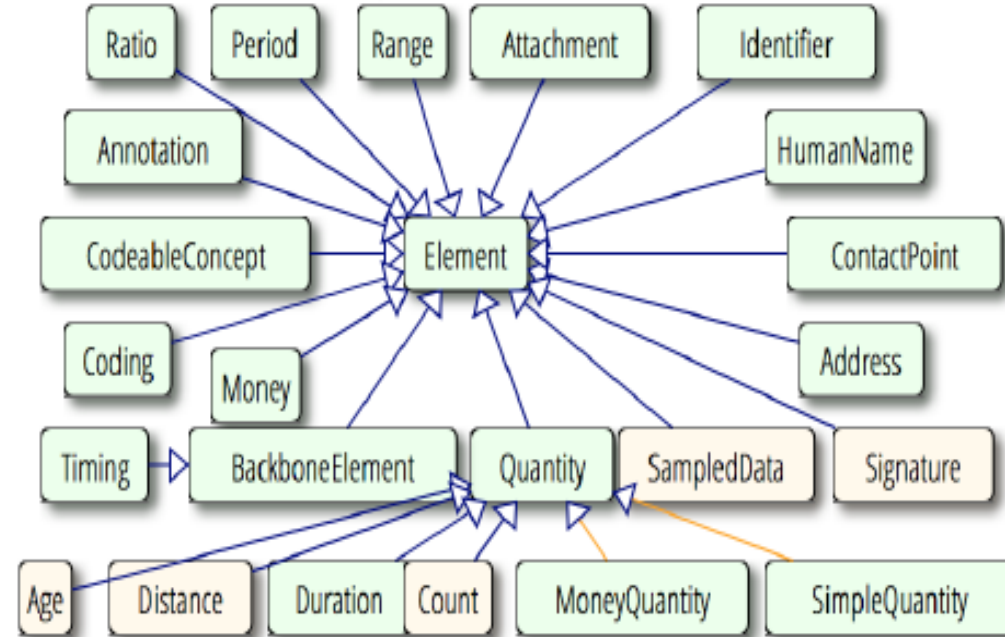
Foundation	Conformance • CapabilityStatement N • StructureDefinition N • ImplementationGuide 1 • SearchParameter 3 • MessageDefinition 1 • OperationDefinition N • CompartmentDefinition 1 • StructureMap 2 • GraphDefinition 1 • ExampleScenario 0	Terminology • CodeSystem N • ValueSet N • ConceptMap 3 • NamingSystem 1 • TerminologyCapabilities 0	Security • Provenance 3 • AuditEvent 3 • Consent 2	Documents • Composition 2 • DocumentManifest 2 • DocumentReference 3 • CatalogEntry 0	Other • Basic 1 • Binary N • Bundle N • Linkage 0 • MessageHeader 4 • OperationOutcome N • Parameters N • Subscription 3
	Individuals • Patient N • Practitioner 3 • PractitionerRole 2 • RelatedPerson 2 • Person 2 • Group 1	Entities #1 • Organization 3 • OrganizationAffiliation 0 • HealthcareService 2 • Endpoint 2 • Location 3	Entities #2 • Substance 2 • BiologicallyDerivedProduct 0 • Device 2 • DeviceMetric 1	Workflow • Task 2 • Appointment 3 • AppointmentResponse 3 • Schedule 3 • Slot 3 • VerificationResult 0	Management • Encounter 2 • EpisodeOfCare 2 • Flag 1 • List 1 • Library 2
	Summary • AllergyIntolerance 3 • AdverseEvent 0	Diagnostics • Observation N • Media 1	Medications • MedicationRequest 3 • MedicationAdministration 2	Care Provision • CarePlan 2 • CareTeam 2	Request & Response • Communication 2 • CommunicationRequest 2

Data Types

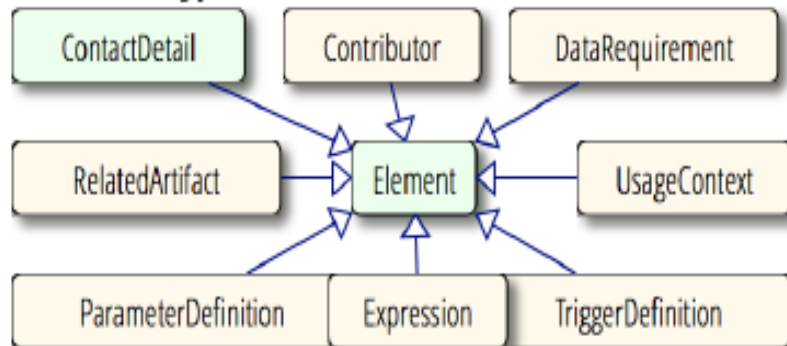
Primitive Types



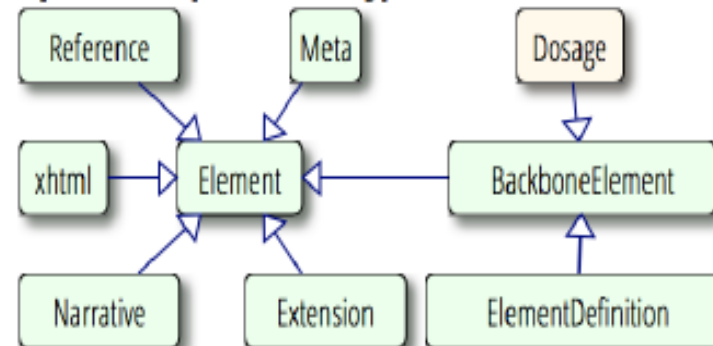
General-Purpose Data types



Metadata Types

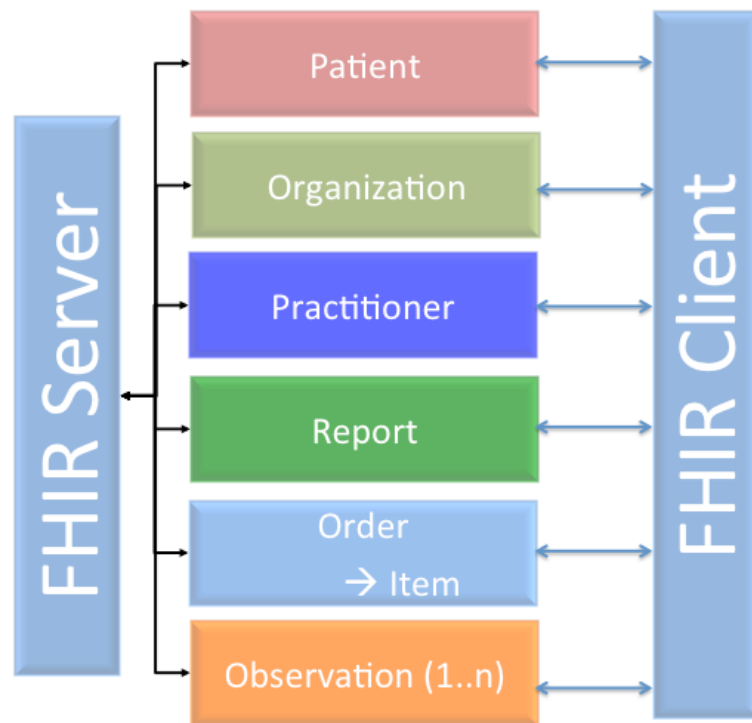


Special Purpose Data types

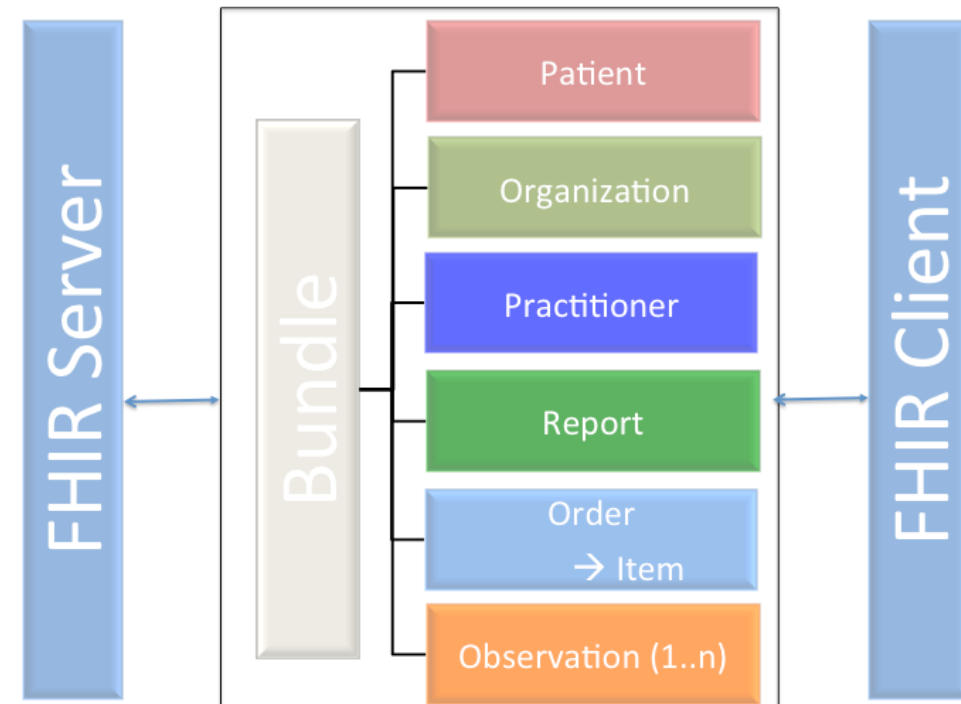


Bundles

- A lab report in FHIR can be represented as a set of related resources: patient, requesting physician, observations, report in PDF format, order.



Individual resources can be transmitted as multiple transactions



Multiple resources bundled for a logical purpose as a transaction

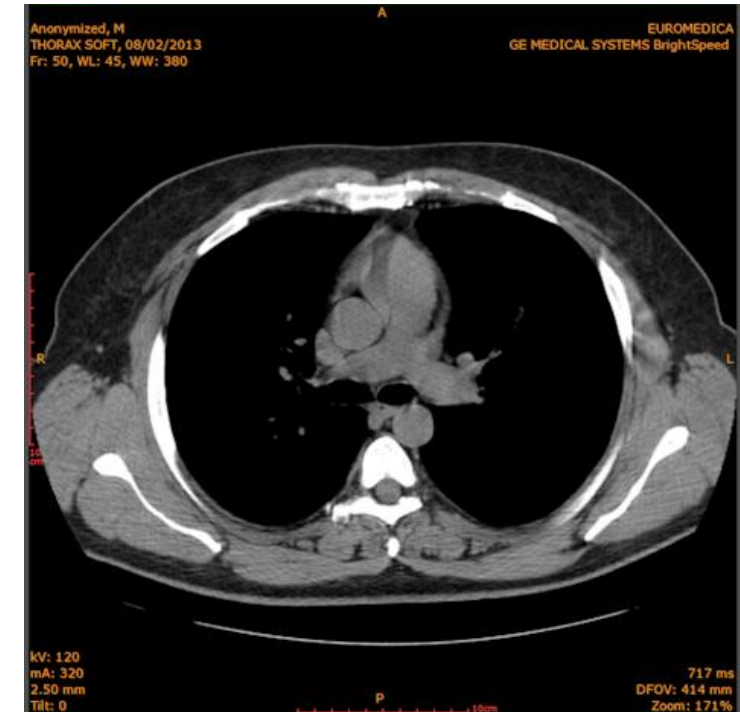


FHIR WITH OTHER STANDARDS

Digital Imaging and Communications in Medicine (DICOM)



- Standard by National Electrical Manufacturers Association (NEMA)
- Global specification for exchanging medical images in radiology, pathology, cardiology and related disciplines
- Majorly used to standardize output of medical imaging hardware (X-Ray, CT, MRI machines) and software (PACS, Radiological Reporting)
- DICOM is hierarchical information structure, has a Client-Server service architecture, and also provides File/DIR specifications
- Using DICOM with FHIR
 - The [DiagnosticReport](#) FHIR Resource can carry a DICOM ImagingStudy while exchanging information

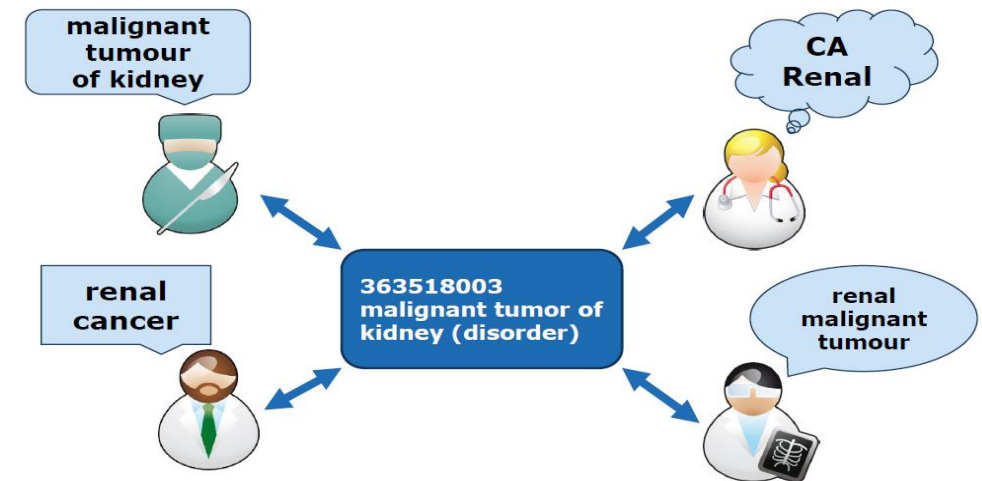


- Maintained and distributed by SNOMED International (also known as IHTSDO), a SDO with about 47 countries as members/owners.
- Structured vocabulary used in clinical practice to accurately describe the care and treatment of patients

- Purpose

- Semantic Interoperability
- To support clinical care:

- Recording statements about the health and health care of an individual patient
- Express meaning at various levels of abstraction for clinicians, patients, researchers or organisations



SNOMED CT Coverage



- Comprehensive, multilingual clinical healthcare terminology
- Covers 18 major healthcare domains with several sub-domains
- Contains **3,60,940 unique concepts** relevant in healthcare, with more than **1.4 million descriptions** and more than two and a half million relationships between them
- Extendible for National Requirements
 - AYUSH medicine
 - Drug codes
 - Any other specific to India
- Useful in research, data analytics, CDSS, etc.

Diagnosis

Symptoms/Chief Complaints

Laboratory

Procedure/ Treatment

Orders, Results

Medications

Body Structure

Nursing

Substance

Medical Devices

Medical History

Organisms

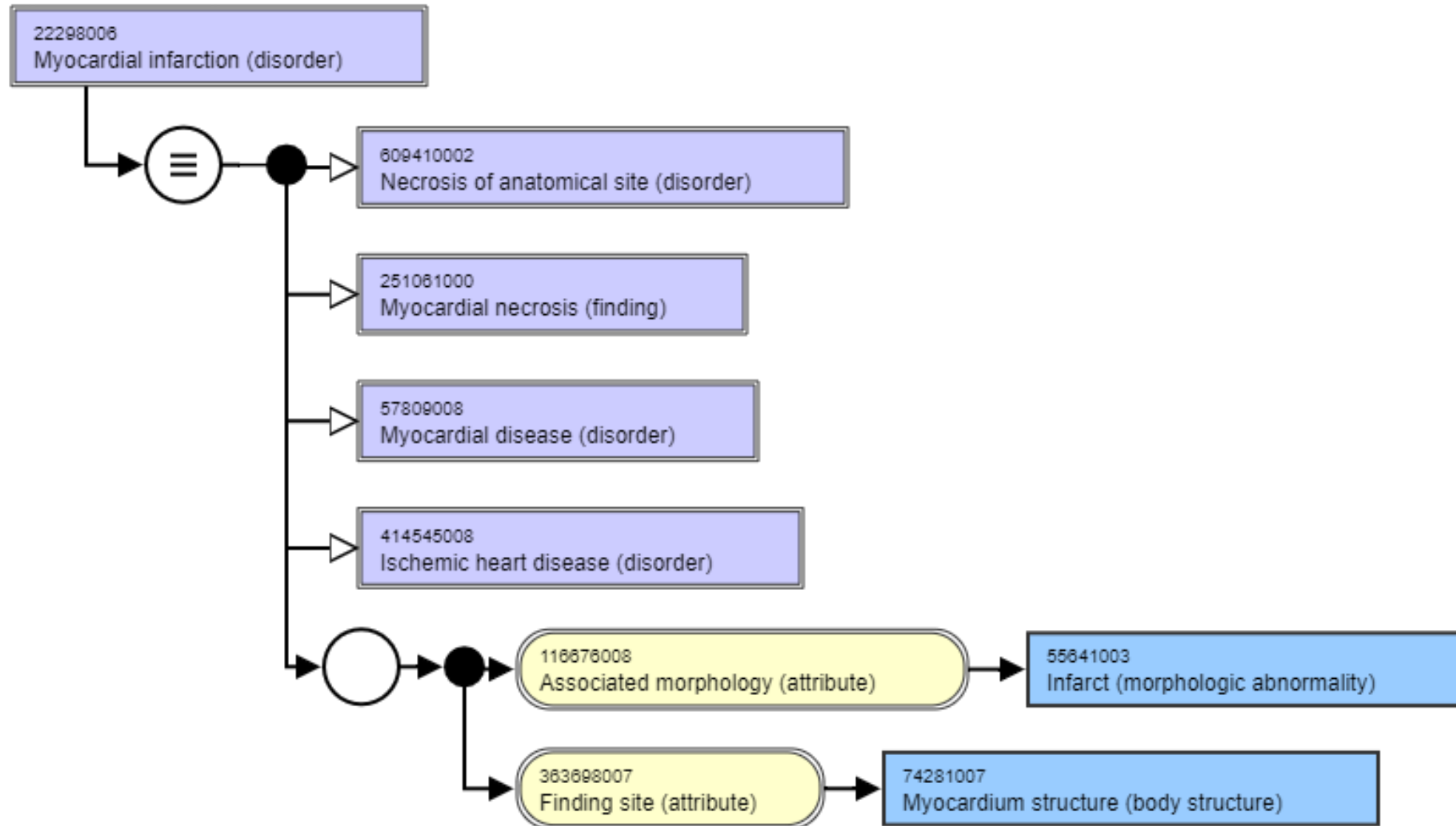
Specimen

Events

Environment / Geographical location

Record artifacts

SNOMED CT Data Model



SNOMED CT in Health Records



History

Active Medicines

Medical Condition

Mr. [REDACTED]

MRN : ZFO260

| DOB Not Available [Add DOB](#)

Last Visited : 13 Oct 2017, 03:41 am

Public

Payment

Prescription

Finish

ALDICORD 16MG TABLET

AZITHIOPRINE 50MG TABLET

URIVOID 25MG TABLET

AZOMARC 250MG TABLET

Lucrey-Driscoll Syndrome

Peripheral Neuralgia

Typhoid Fever

Common Migraine

Gingivitis

e-Prescription

Load Previous

Clear

Symptom

Vitals

Symptoms

Disorders

Clinical Notes

Chest pain

+ Add Findings / Symptoms

Acidaemia

Common migraine

+ Add Disorders / Disease

Patient complaining of uneasiness and acidity

पुरी नींद ले , हल्के प्रानयाम करे

Type Here...

G

EN

Q Search Symptom

Foot joint temperature finding (finding)

Expand

Peripheral neuropathic pain

Expand

O/E - cold and clammy

Expand

Malarial fever

Expand

Red eye

Expand

Portal-systemic shunt

Expand

Ankle pain (finding)

Expand

Chill (finding)

Expand

Fever

Expand

Swollen gums (finding)

Expand

Excessively deep breathing

Expand

Temporomandibular joint tenderness on palpation

Expand

Malignant tertian fever (finding)

Expand

SNOMED CT in FHIR



```
{
  "resourceType": "Condition",
  "id": "example-01",
  "meta": {
    "profile": [
      "https://nrce.in/ndhm/fhir/r4/StructureDefinition/Condition"
    ]
  },
  "text": {
    "status": "generated",
    "div": "<div xmlns=\\"http://www.w3.org/1999/xhtml\\">Abdominal pain on 09-July 2020</div>"
  },
  "code": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "21522001",
        "display": "Abdominal pain"
      }
    ]
  },
  "text": "Abdominal pain"
},
"subject": {
  "reference": "Patient/1"
}
}
```

```
  "text": {
    "status": "generated",
    "div": "<div xmlns=\\"http://www.w3.org/1999/xhtml\\">Placement of stent in coronary artery</div>"
  },
  "status": "completed",
  "code": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "36969009",
        "display": "Placement of stent in coronary artery"
      }
    ]
  },
  "text": "Placement of stent in coronary artery"
},
"subject": {
  "reference": "Patient/1"
},
}
```

Logical Observation Identifiers Names and Codes (LOINC)



- Universal standard for identifying (coding) medical and laboratory observations
- Created and is maintained by the Regenstrief Institute, a US nonprofit medical research organization.
- Essentially a laboratory coding system
- Details the method, components, system, etc. used for doing a particular test / observation
 - 9156-0 *Glucose [Moles/volume] in Urine by Automated test strip*
 - Glucose:SCnc:Pt:Urine:Qn:Test strip

LOINC with FHIR



```
"code" : {  
  "coding" : [  
    {  
      "system" : "http://loinc.org",  
      "code" : "35200-5",  
      "display" : "Cholesterol [Moles/volume] in Serum or Plasma"  
    }  
  ],  
  "text" : "Cholesterol"  
},
```

LOINC Panel (Logical group of LOINC test observations)

Individual LOINC Test Code

```
"code" : {  
  "coding" : [  
    {  
      "system" : "http://loinc.org",  
      "code" : "24331-1",  
      "display" : "Lipid 1996 panel - Serum or Plasma"  
    }  
  ],  
  "text" : "Lipid Panel"  
},
```

International Classification of Diseases (ICD)



- Created and Maintained by the World Health Organization (WHO)
- Designed as a health care classification system, providing a system of **diagnostic codes** for classifying diseases
- Supported clinical care use cases:
 - Monitoring of the incidence and prevalence of a disease
 - Observing reimbursements and resource allocation trends
 - Keeping track of safety and quality guidelines
- Enable aggregation of data to support their decisions and policies
- Allows government bodies to conduct epidemiological research of health trends
- Allows health data comparisons in the same location across different time periods

Acute upper respiratory infections (J00-J06)

Excl.: chronic obstructive pulmonary disease with acute exacerbation NOS ([J44.1](#))

J00 Acute nasopharyngitis [common cold]

Incl.: Coryza (acute)
Nasal catarrh, acute
Nasopharyngitis:
• NOS
• infective NOS
Rhinitis:
• acute
• infective

Excl.: nasopharyngitis, chronic ([J31.1](#))
pharyngitis:
• NOS ([J02.9](#))
• acute ([J02.-](#))
• chronic ([J31.2](#))
rhinitis:
• NOS ([J31.0](#))
• allergic ([J30.1-J30.4](#))
• chronic ([J31.0](#))
• vasomotor ([J30.0](#))
sore throat:
• NOS ([J02.9](#))
• acute ([J02.-](#))
• chronic ([J31.2](#))

J01 Acute sinusitis

Incl.: abscess
empyema
infection
inflammation
suppuration
acute, of sinus (accessory)(nasal)

Use additional code (B95-B98), if desired, to identify infectious agent.

Excl.: sinusitis, chronic or NOS ([J32.-](#))

J01.0 Acute maxillary sinusitis

Acute antritis

J01.1 Acute frontal sinusitis

J01.2 Acute ethmoidal sinusitis

J01.3 Acute sphenoidal sinusitis

J01.4 Acute pansinusitis

J01.8 Other acute sinusitis

Acute sinusitis involving more than one sinus but not pansinusitis

J01.9 Acute sinusitis, unspecified

ICD-10 with Other Standards



- ICD-10 with SNOMED CT
 - The granular data entry happens in SNOMED CT
 - Standard SNOMED CT to ICD-10 mapping is available
 - ICD-10 code based classification is reported for epidemiology, insurance, etc.
- ICD-10 with FHIR
 - The Claims Resource links to ICD-10 code

diagnosis	0..*	BackboneElement	Missing Tooth Reason Codes (Example) Pertinent diagnosis information
sequence	1..1	positiveInt	Diagnosis instance identifier
diagnosis[x]	1..1		Nature of illness or problem ICD-10 Codes (Example)
diagnosisCodeableConcept		CodeableConcept	
diagnosisReference		Reference(Condition)	
type	0..*	CodeableConcept	Timing or nature of the diagnosis Example Diagnosis Type Codes (Example)
onAdmission	0..1	CodeableConcept	Present on admission Example Diagnosis on Admission Codes (Example)
packageCode	0..1	CodeableConcept	Package billing code Example Diagnosis Related Group Codes (Example)



ADOPTION OF FHIR IN ABDM

Approach used for Data sharing in NDHM



Fast Healthcare Interoperability Resource (FHIR) standard based data sharing for all HI Types supporting data in 03 formats -

- Scanned Documents
 - Helps uploading historic data
 - Allows quick on-boarding
- Structured data without standard terminology
 - Structured data sharing with flexibility to include free text
 - Better understanding of context
 - Tools like Natural Language Processing can be used for analytics in future
- Fully structured data with Coding / Terminology (SNOMED CT / LOINC/ ICD)
 - Better understanding of data (interoperable and processable data)
 - Data accuracy is guaranteed to the maximum
 - Data ready for analytics which can help in real time decision support
- DICOM for medical images



```
],  
  "text" : "Abdominal pain"  
},  
"subject" : {  
  "reference" : "Patient/1"  
}  
}
```

```
"code" : {  
  "coding" : [  
    {  
      "system" : "http://snomed.info/sct",  
      "code" : "36969009",  
      "display" : "Placement of stent in coronary artery"  
    }  
  ],  
  "text" : "Placement of stent in coronary artery"  
}
```

FHIR Implementation Guide Build 3.0.0



- FHIR Implementation Guide (IG) for ABDM:
<https://nrces.in/ndhm>
- Describes requirements for ABDM FHIR implementation
 - HL7 FHIR, DICOM, ICD-10, SNOMED CT, LOINC
- **50 FHIR profiles** include **07 Clinical Artifacts** (HI Types); and 39 core profiles
 1. OP Consultation Note
 2. Discharge Summary
 3. Prescription
 4. Diagnostic Report
 5. Wellness Record
 6. Immunization Record
 7. Health Record Document
- **04 Code Systems** and **26 Value sets** from standard clinical terminologies to support their use in profiles

FHIR Implementation Guide for ABDM
3.0.0 - CI Build

Home Profiles HCX Profiles Extensions Terminology Examples Downloads

FHIR Profiles for
ABDM Health Data Interchange Specifications

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FHIR Implementation Guide for ABDM - Local Development build (v3.0.0). See the [Directory of published versions](#)

1 Home

Official URL: https://nrces.in/ndhm/fhir/r4/ImplementationGuide/ndhm.in	Version: 3.0.0
Active as of 2023-04-20	Computable Name: FHIR-Implementation-Guide-for-ABDM

1.1 Background

Enduring with the vision of National Health Policy (NHP) 2017, 'Health and wellbeing for all at all ages', Ministry of Health and Family Welfare (MoHFW), Government of India recognized the need for creating a framework for the evolution of a National Digital Health Eco-system (NDHE), which will support 'Continuum of care' for an individual.

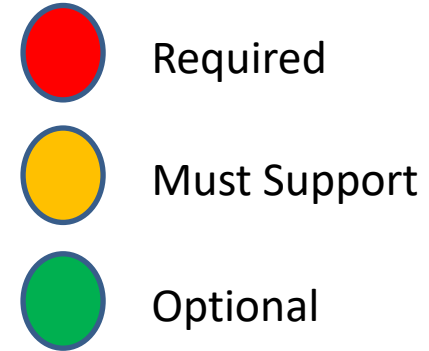
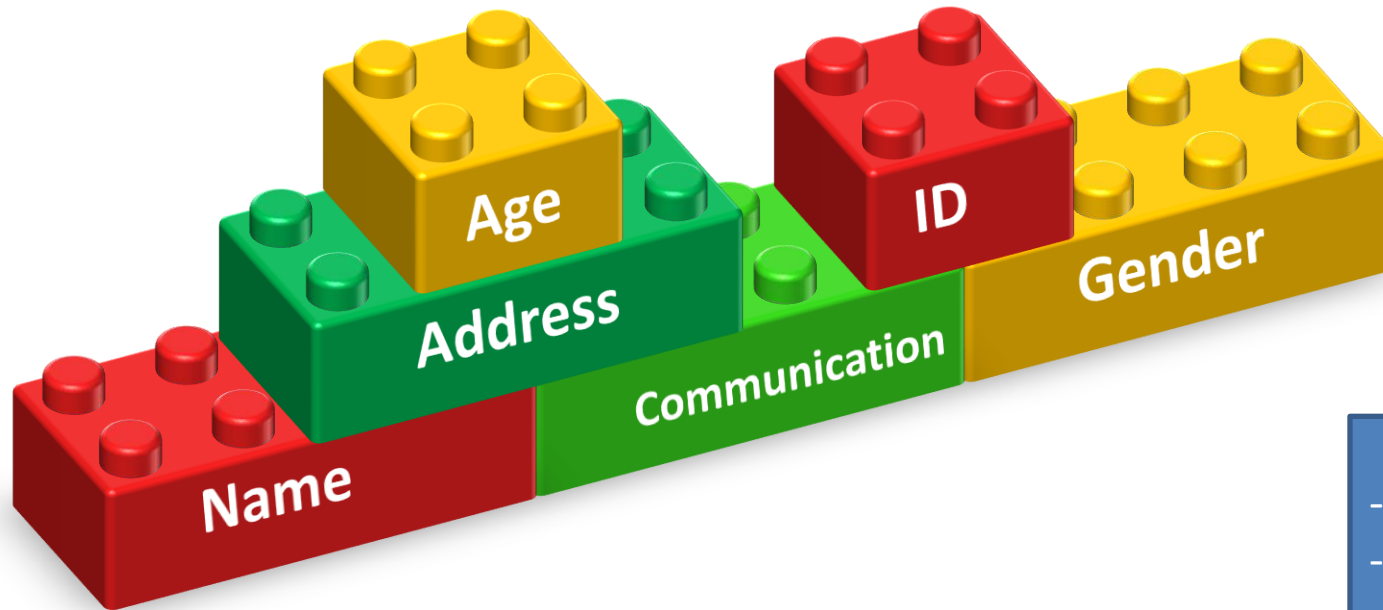
To create and enable digital health ecosystem and prioritize digital health in India, and to develop an implementation framework for the National Health Stack, the committee constituted by MoHFW Government of India produced the [National Digital Health Blueprint \(NDHB\)](#), laying out the building blocks and an action plan to comprehensively and holistically implement digital health.

The Government has established the [Aarushman Bharat Digital Mission \(ABDM\)](#) with The Ministry of Health and Family Welfare defining the policy and

- Background
- Introduction
- Purpose and Scope
- ABDM Actors
- How to read this Guide
- ABDM Profiles

[www.nrces.in/ndhm](https://nrces.in/ndhm)

- FHIR specification satisfy the majority of common use cases
 - Provides flexibility to extend and adapt it according to local needs
 - Almost every element in the base FHIR specification is optional
- The conformance on content of the data and operations is needed
- FHIR profiling is a way to build consensus on content, operations & search parameters supported
 - Defining required and optional resource elements
 - Defining additional resource elements are needed
 - Binding terminology and value sets to be used
 - Deciding which API features are used, and how



Patient Resource

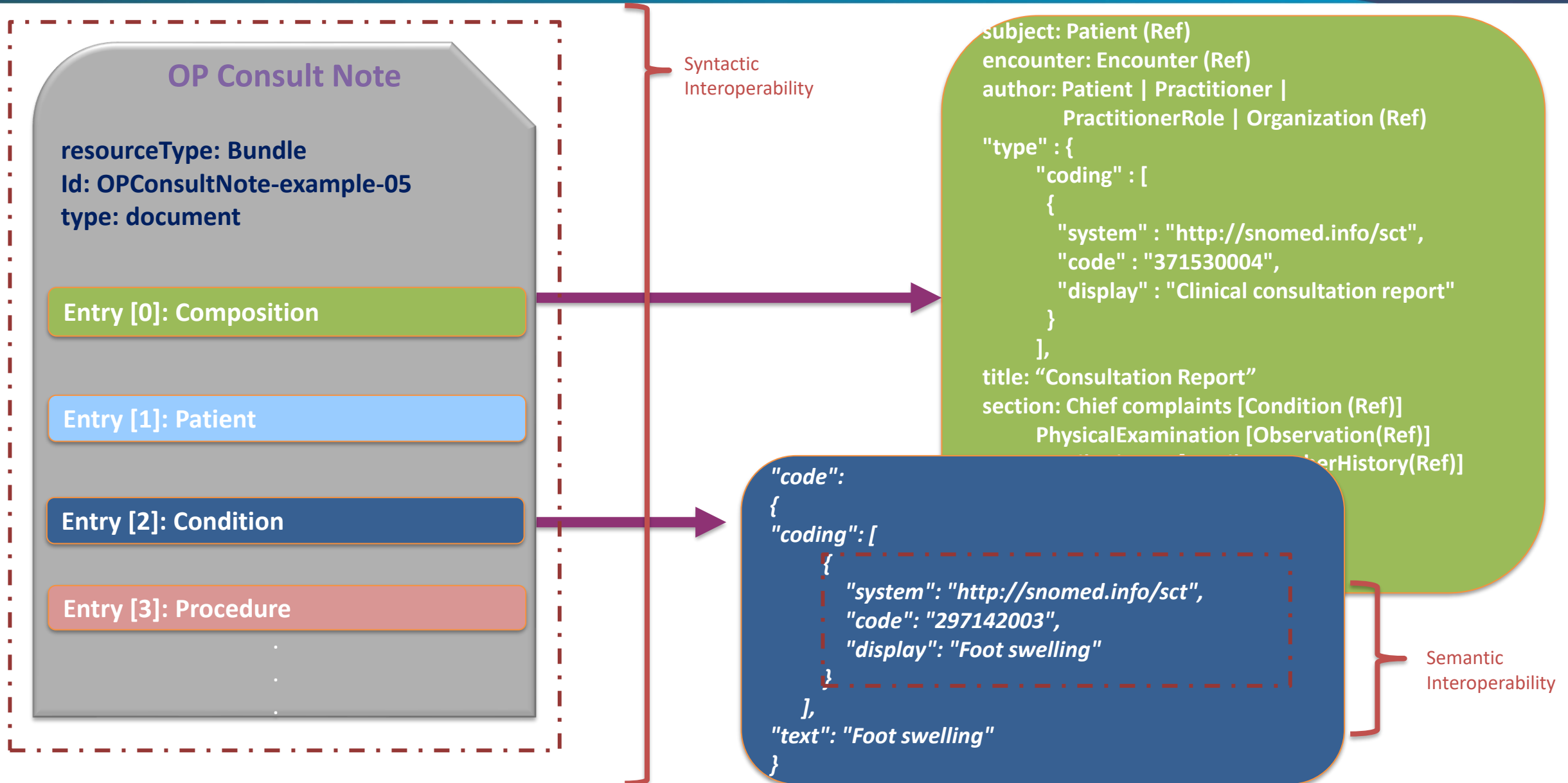
- Define what is required, optional, etc.
- Allow building light weight objects
- Enable Minimal Data sharing
- Provision for maximum data sharing (as needed in different contexts)
- Reuse standard defined data structures
- Productive time investment

Reading FHIR Profiles for ABDM



- **Text Summary:** provides human-readable summary of changes that we have made. **It refers to the Differential View of a profile.**
- **Differential View:** mentions the elements that are changed while profiling
- **Mandatory Element:** Element with cardinality 1..1, 1..*
-  : element that must be supported.
- **MUST Support:** implementations that produce or consume resources SHALL provide "support" for the element in some meaningful way
- **MUST Support for ABDM:** The Receiver (HIU) MUST have the capacity to read the element though it is optional. The Sender (HIP) may opt not to fill the element while sending (when data is not available)

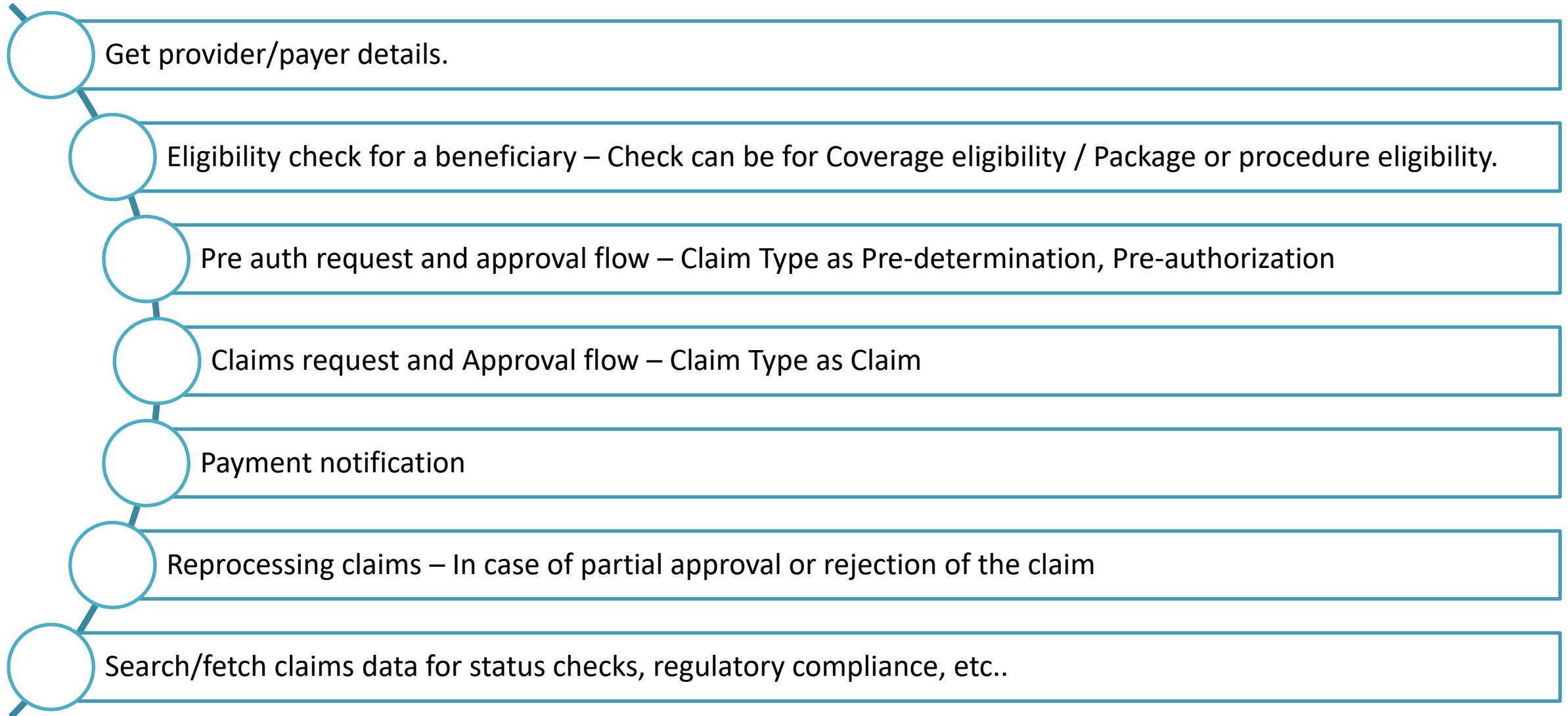
Understanding the Structure of ABDM HI Types in FHIR





ADOPTION OF FHIR IN NHCX

Key Use cases



Profiling Requirements for NHCX

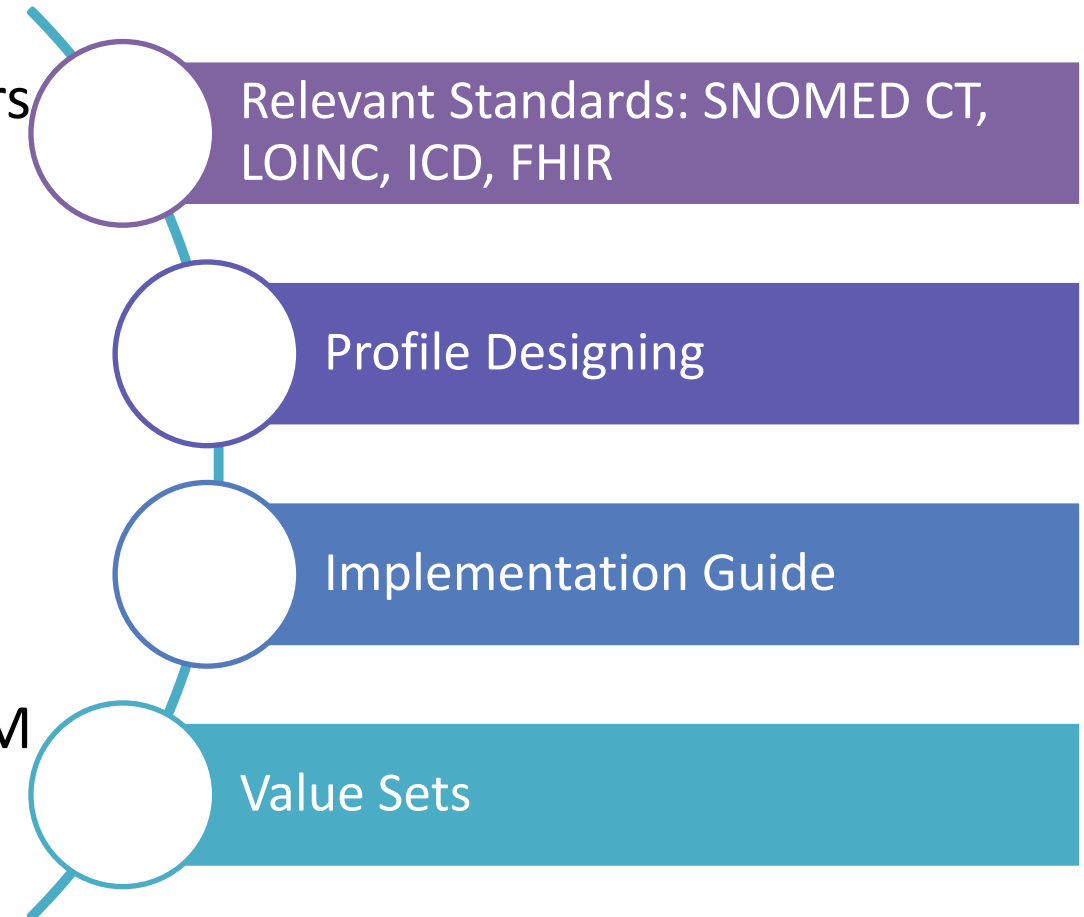


- Ensure the *cycle specific resources* are being shared
- Ease of use, validation and processing
- FHIR resource structures to be adopted over NHCX APIs
- The profiling should be generic enough to support any insurance workflow (including PMJAY)
- Value sets from Indian context are to be defined
- Ensure the practical scenarios of insurance processing are covered from Indian context (Proof of Presence, Proof of Identity, etc.)
- Leverage the ABDM profiles & Resources and extend the same IG for NHCX

NHCX Standard Development Approach



- Collaborative approach
 - End-user driven
 - Include Public & Private communities/users
 - Consultation and feedback from national agencies (NHA, IRDIA, etc.)
- Focus on implementing best suitable design
 - Use-case driven
- Re-use existing efforts (ABDM profiles)
- Leverage the existing/community work
- NRCeS to be the SPC and Maintainer for ABDM Profiles, NHCX Profiles like CDCl, AYUSH standards and other initiatives of MoHFW



FHIR Resource Under Consideration for NHCX



- **InsurancePlan**: InsurancePlan represents the product that can be offered to the customer. It describes a health insurance offering comprised of a list of covered benefits, costs associated with those, and additional information about the offering.
- **Claim**: Claims are used to exchange financial and clinical information between healthcare providers, payors /insurers.
- **Claim Response**: Claim Response resource provides application-level adjudication results or errors after processing a submitted Claim.
- **Coverage Eligibility Request**: The Coverage Eligibility Request asks the insurer for a CoverageEligibility Response with information about valid coverage, available coverages, provided benefits, and pre-auth requirements.
- **Coverage Eligibility Response**: Coverage Eligibility Response provides eligibility and plan details from a Coverage Eligibility Request.



- FHIR profiles NHCX

- Health Claim Artifacts

Following artifact are profile based on [Bundle](#) resource of FHIR R4.0.1:

Name	Based On	Definition
ClaimBundle	Bundle	The profile is based on a Bundle of type collection, where all the supporting information required for processing claim can be shared. Multiple entries can be added in a bundle to provide information like financial, clinical, provision of health care services with payors and for reporting to regulatory bodies and firms which provide data analytics. The bundle can be generated depending on the nature of the request defined by 'use' element in a claim resource, like preauthorization, predetermination and claim and can be shared over NHCX ecosystem.
ClaimResponseBundle	Bundle	The profile is based on a Bundle of type collection, where adjudicated response to a Claim, Predetermination or Preauthorization related information is carried. Multiple entries included in a bundle carries the information and provides application level adjudication results.
CoverageEligibilityRequestBundle	Bundle	The profile is based on a Bundle of type collection, where all the information required to process Coverage Eligibility Request can be shared. Depending on the purpose of the request like validation, discovery,auth-requirement and benefit data can be included in the etries of a bundle.
CoverageEligibilityResponseBundle	Bundle	The Coverage Eligibility Response Bundle is a Bundle profile with type collection. The bundle profile provides the response and plan details from the processing of an CoverageEligibilityRequest resource.

[InsurancePlanBundle](#)

5.2 ValueSets

These value sets have been defined for this implementation guide.

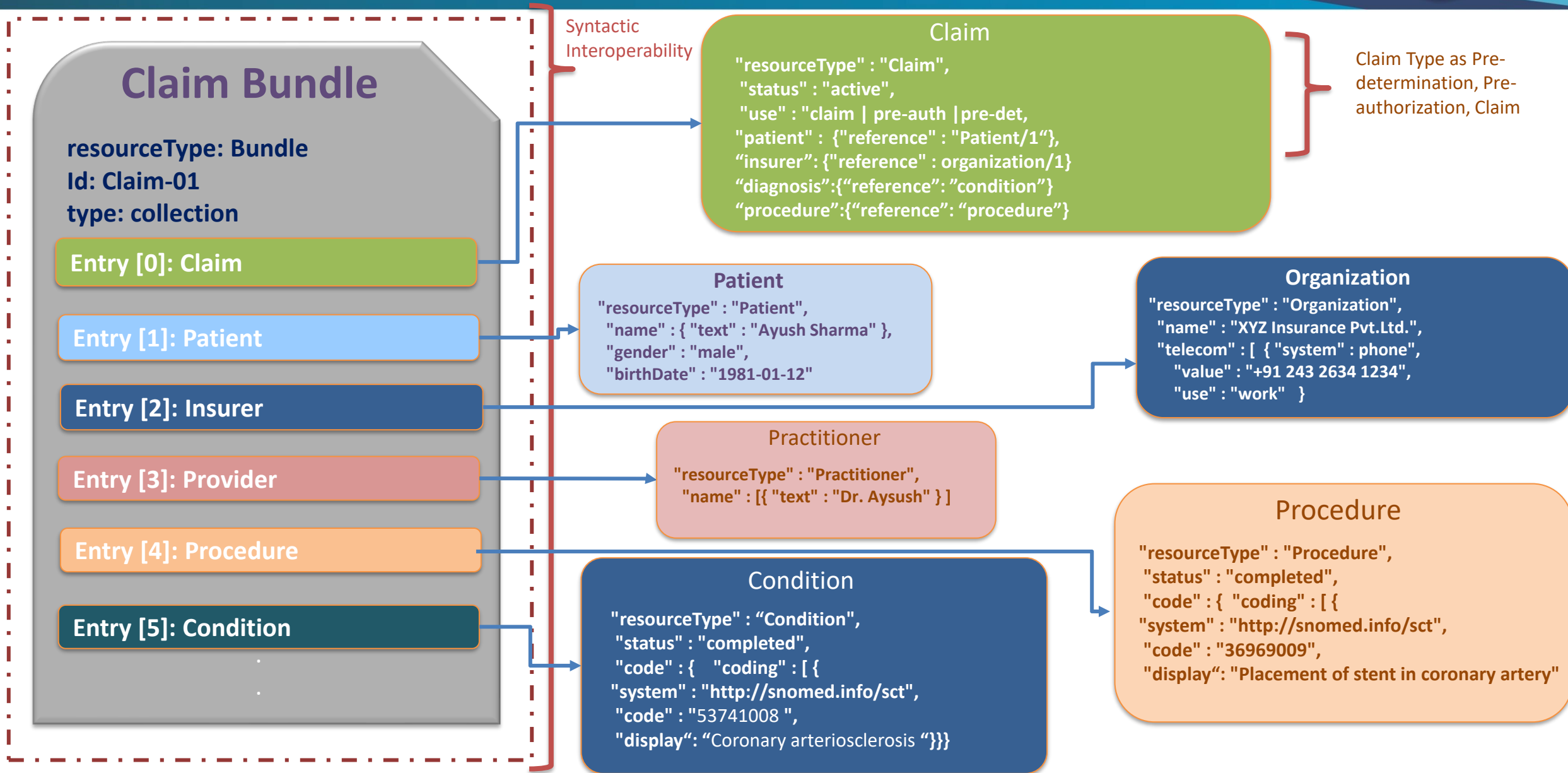
- Administrative ValueSets

Name	Definition
Benefit Category	This Valueset includes a collection of SNOMEDCT codes related to BenefitCategory.
Benefit Type	This ValueSet includes a collection of codes that can be used to categorize the type of benefits provided by an insurance.
Care Team Qualification	This valueset comprises codes that represent credentials or specializations of healthcare practitioners.
Care Team Role	This valueset comprises codes that represent various roles within a care team.
Claim Type	This ValueSet contains a set of codes that can be utilized to state the type of Claim.
ChargeItem Types	This Valueset contains codes that represent the various types of items being charged for in invoice.
Coverage Type	This ValueSet consists of a set of codes that can be utilized to categorize the insurance coverage type.
Diagnostic Code	This valueset consists of codes that represent ICD-10 Diagnostic codes.
Diagnostic Type	This valueset contains codes that represent the various types of diagnoses.
Invoice Types	This Valueset contains codes that represent the various types of invoices.
Identifier Type Code	This Valueset contains coded type for an identifier that can be used to determine which identifier to use for a specific purpose

- Relevant Value sets

Preview - <https://nrcea.in/preview/ndhm/fhir/r4/hcx-profile.html>

Understanding the Structure of Claim Bundle



- FHIR Implementation Guide (IG) for ABDM: <https://nrces.in/ndhm>
- Describes requirements for ABDM FHIR implementation
 - HL7 FHIR, DICOM, ICD-10, SNOMED CT, LOINC
- **50 FHIR profiles** include **07 Clinical Artifacts** (HI Types); **04 Health Claim Exchange (HCX)** profiles and 39 core profiles
 1. Claim Bundle
 2. Claim Response Bundle
 3. Coverage Eligibility Request Bundle
 4. Coverage Eligibility Response Bundle
- Claim
- Claim Response
- Eligibility Request
- Eligibility Response
- Insurance Plan
- Coverage
- Payment Notice
- Payment Reconciliation
- Task

Examples



```
{
  "resourceType" : "CoverageEligibilityRequest",
  "id" : "preauth-example-01",
  "meta" : {
    "profile" : [
      "https://nrce.in/ndhm/fhir/r4/StructureDefinition/CoverageEligibilityRequest"
    ]
  },
  "text" : {
    "status" : "generated",
    "div" : "<div xmlns='http://www.w3.org/1999/xhtml'><p><b>Generated Narrative: CoverageEligibilityRequest</b></p></div>"
  },
  "identifier" : [
    {
      "value" : "9900"
    }
  ],
  "status" : "active",
  "priority" : {
    "coding" : [
      {
        "code" : "normal"
      }
    ]
  },
  "productOrService" : {
    "coding" : [
      {
        "system" : "http://snomed.info/sct",
        "code" : "10849003",
        "display" : "Removal of foreign body"
      }
    ]
  },
  "diagnosis" : [
    {
      "id" : "1111",
      "diagnosisReference" : {
        "reference" : "Condition/example-01"
      }
    }
  ],
  "purpose" : [
    "validation",
    "auth-requirements"
  ]
}
```

CoverageEligibilityRequest with purpose: 'auth-requirements'

```
{
  "resourceType" : "Claim",
  "id" : "example-01",
  "meta" : {
    "profile" : [
      "https://nrce.in/ndhm/fhir/r4/StructureDefinition/Claim"
    ]
  },
  "text" : {
    "status" : "generated",
    "div" : "<div xmlns='http://www.w3.org/1999/xhtml'><p><b>Generated Narrative: Claim</b></p></div>"
  },
  "identifier" : [
    {
      "system" : "http://happypharma.com/claim",
      "value" : "7612345"
    },
    {
      "sequence" : 1,
      "diagnosisCodeableConcept" : {
        "coding" : [
          {
            "code" : "654456"
          }
        ]
      }
    }
  ],
  "status" : "active",
  "type" : {
    "coding" : [
      {
        "system" : "http://terminology.hl7.org/CodeSystem/claim-type",
        "code" : "pharmaceutical"
      }
    ]
  },
  "use" : "claim",
  "diagnosis" : [
    {
      "coding" : [
        {
          "system" : "http://terminology.hl7.org/CodeSystem/claim-type",
          "code" : "oral"
        }
      ]
    }
  ]
}
```

Claim with use: 'claim'



Thank You

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